Visual Analysis Report

BOI Mule Account Classification

Problem Statement 2

Dataset: 9,082 accounts | 3,925 features | 81 suspicious

14 Visualizations with Analysis

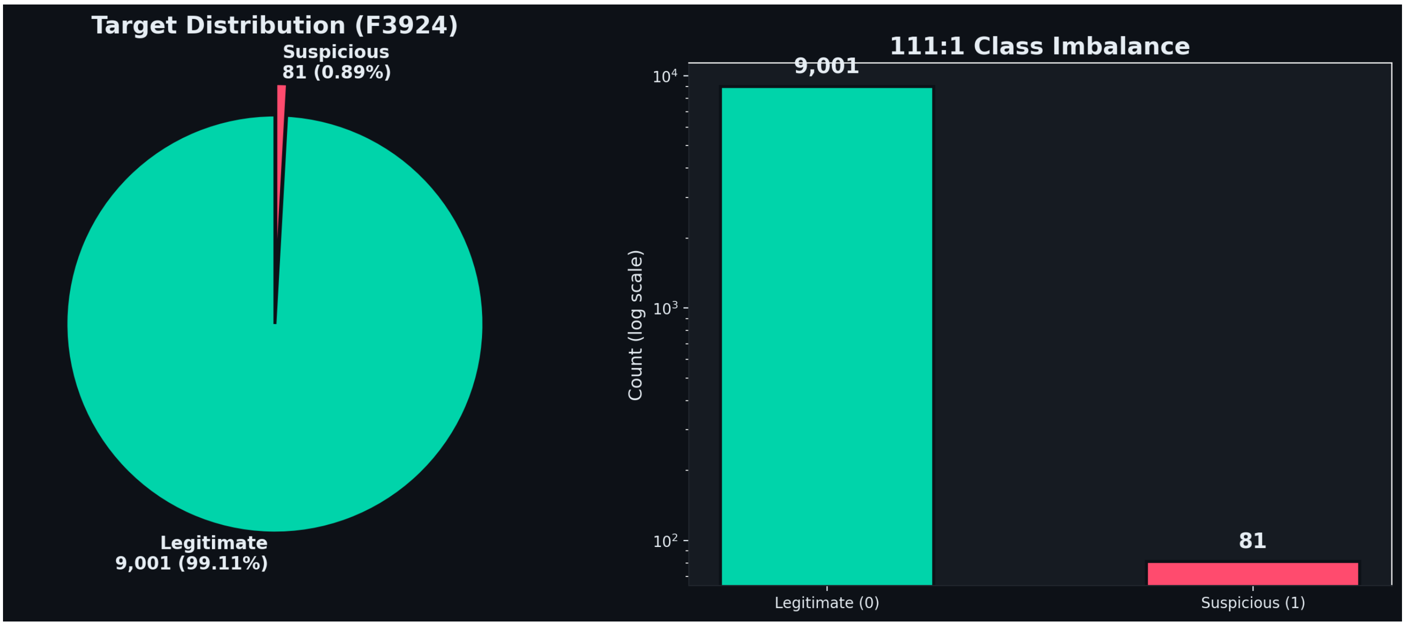
PSBs Cybersecurity, Fraud & AI Hackathon 2026

Bank of India in collaboration with IIT Hyderabad

Powered by DFS (Ministry of Finance) / IBA

June 2026

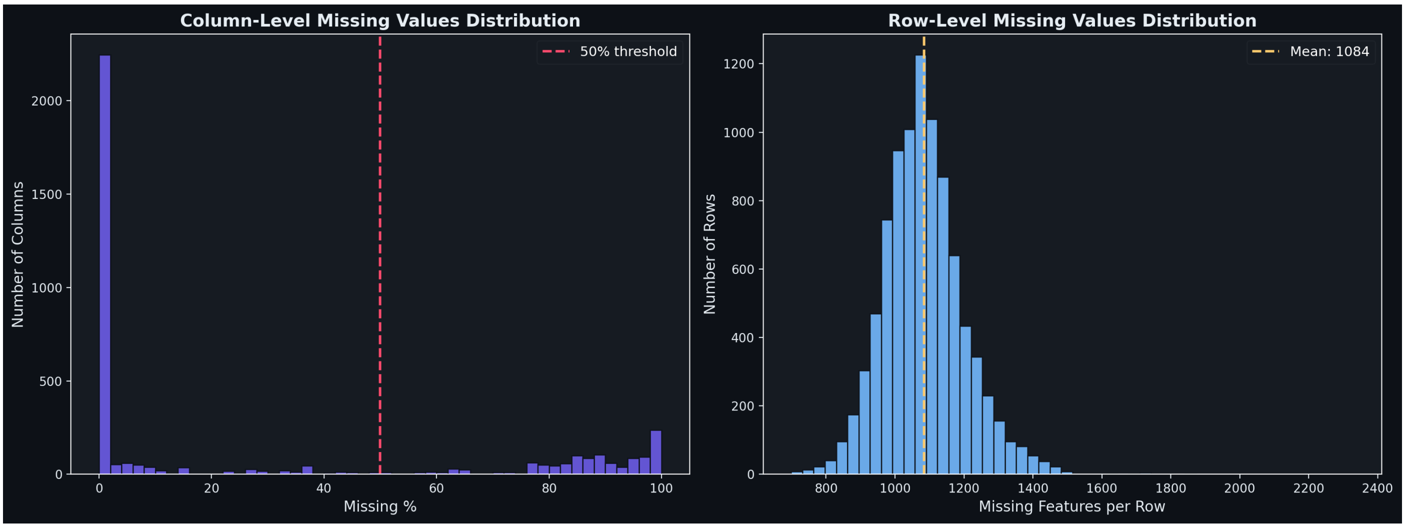
1. Class Imbalance



CAUTION: 111:1 imbalance - Only 81 suspicious accounts out of 9,082. A naive model predicting "legitimate" for everything would score 99.11% accuracy but catch zero fraud.

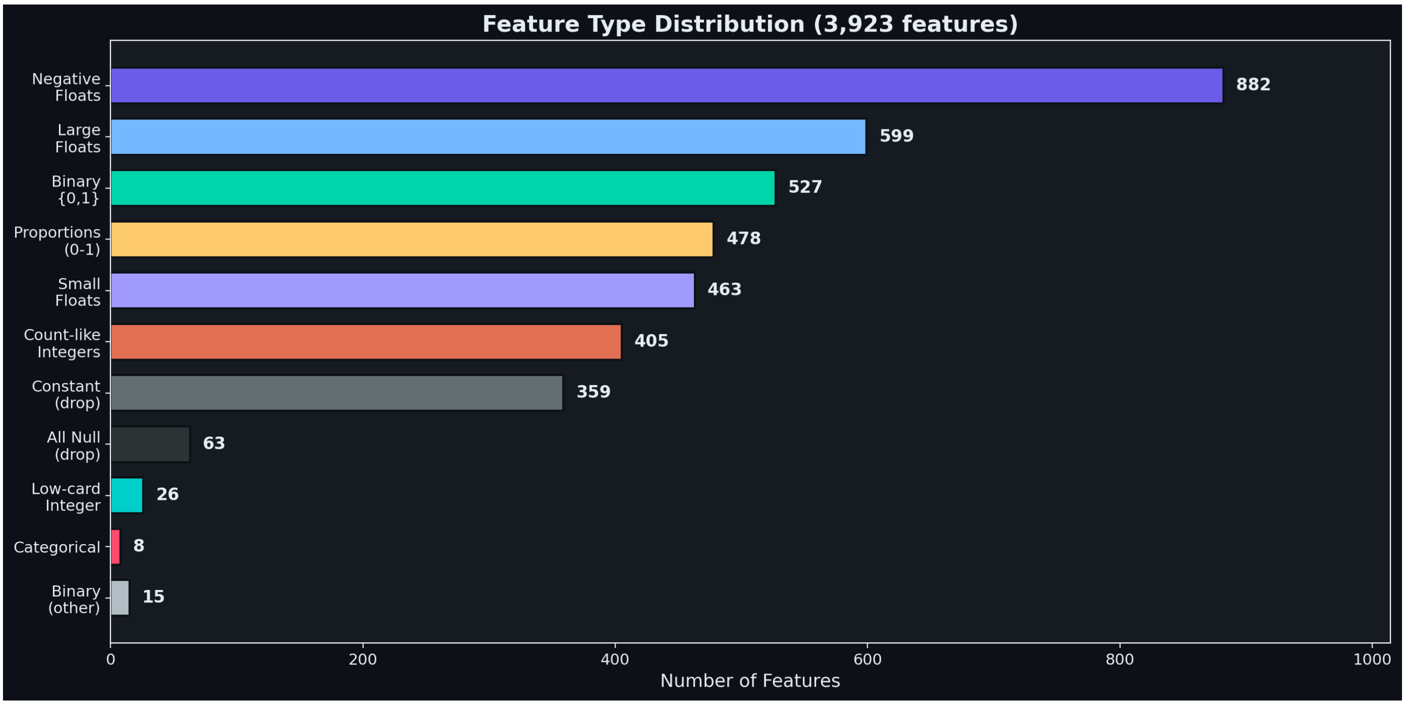
- Must use **SMOTE/ADASYN** or **class weights** during training
- Accuracy is a meaningless metric - use **F1-score** and **PR-AUC** instead
- Stratified splitting is mandatory for cross-validation

2. Missing Values Distribution



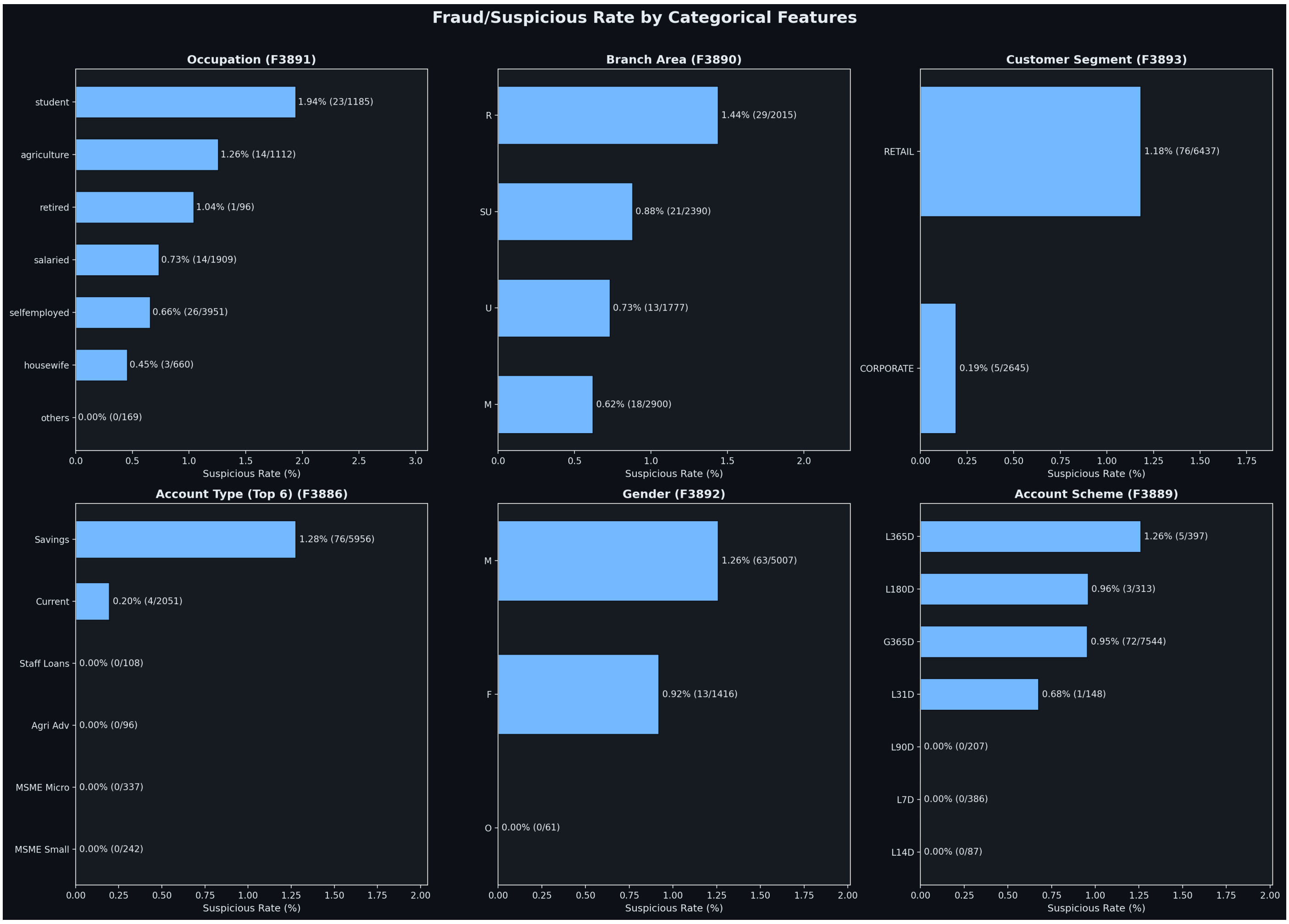
- Left: Most columns with missing data are missing for **10-30%** of rows
- Right: Every row is missing ~**1,084 features** (27.6%) **-** this is structural, not random
- Bimodal pattern suggests **product-dependent features** (e.g., loan features null for non-loan accounts)

3. Feature Type Distribution



- **882 negative-value features** (largest group) **-** delta/change metrics, critical for detecting behavioral shifts
- **527 binary features -** likely product/channel flags
- **478 proportion features** (0-1 range) - behavioral scores and ratios
- **422 features to drop** (359 constant + 63 all-null) - zero predictive value

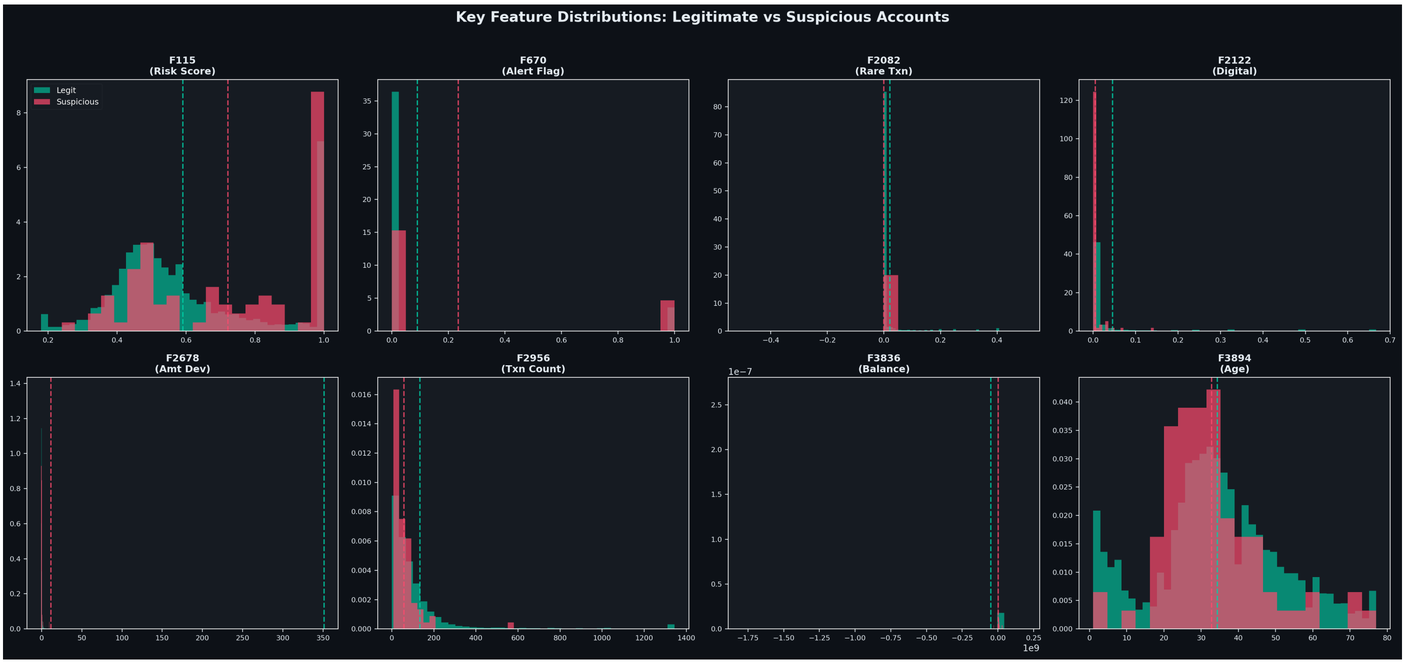
4. Categorical Features vs Fraud Rate



IMPORTANT: Students have the highest fraud rate at 1.94% - nearly 3x the dataset average of 0.89%.

- **Occupation:** Students (1.94%) > Agriculture (1.26%) > Retired (1.04%) > Salaried (0.73%)
- **Branch Area:** Rural (1.44%) is **2.3x** Metro (0.62%)
- **Customer Segment:** RETAIL (1.18%) is **6.2x** CORPORATE (0.19%)
- **Account Type:** Savings (1.28%) dominates - 76 of 81 suspicious accounts
- **Gender:** Males (1.26%) are overrepresented

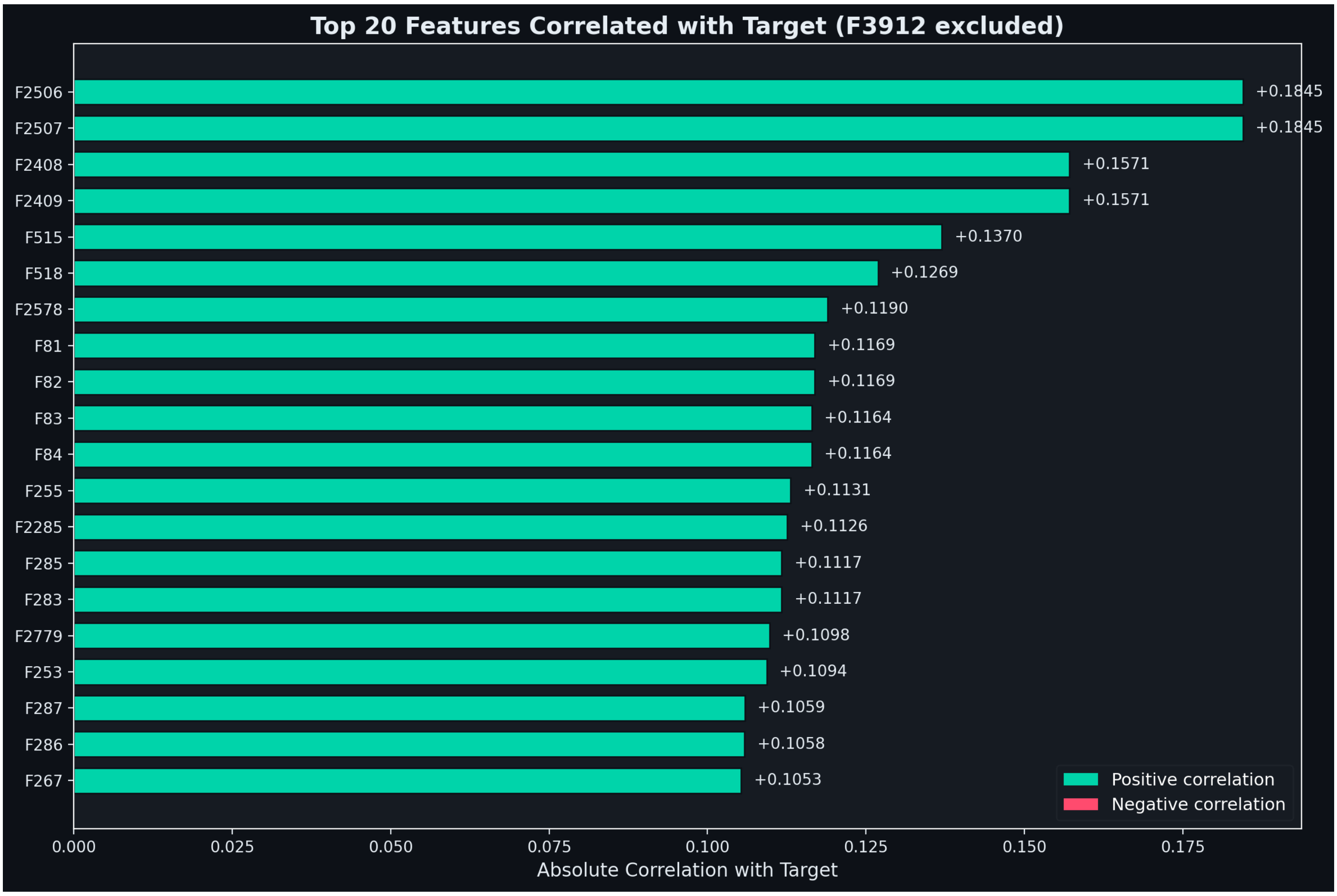
5. Key Features: Legit vs Suspicious



TIP: Dashed vertical lines show class means. Where the red (suspicious) and teal (legit) lines separate, we have signal.

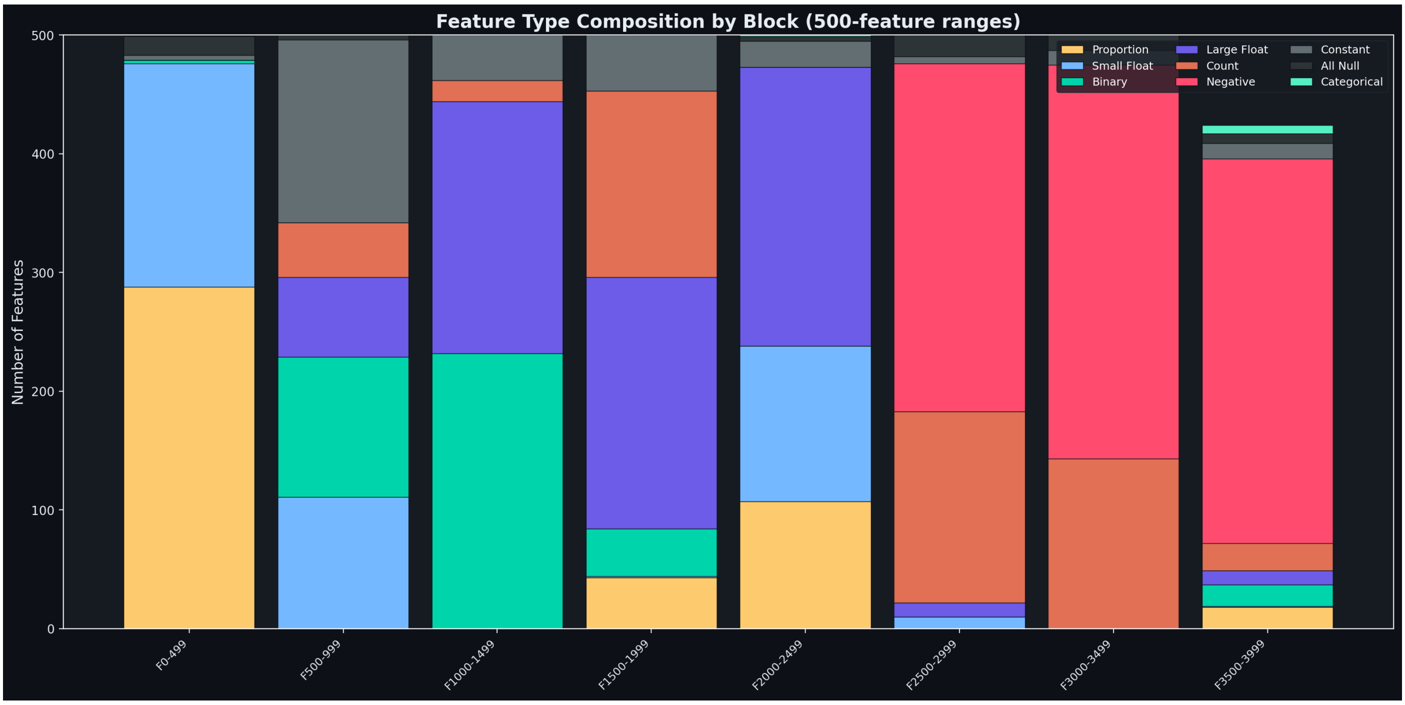
- **F115 (Risk Score):** Suspicious cluster near 0.8, legit spread 0.3-0.7
- **F670 (Alert Flag):** Binary - suspicious have the flag set 2.6x more often
- **F2082 (Rare Txn):** Suspicious show almost **zero** rare transaction activity
- **F2122 (Digital):** Suspicious have near-zero digital channel usage
- **F2956 (Txn Count):** Suspicious cluster at very low counts
- **F3894 (Age):** Suspicious skew **younger** (peak at 20-35)

6. Top 20 Correlated Features (F3912 excluded)



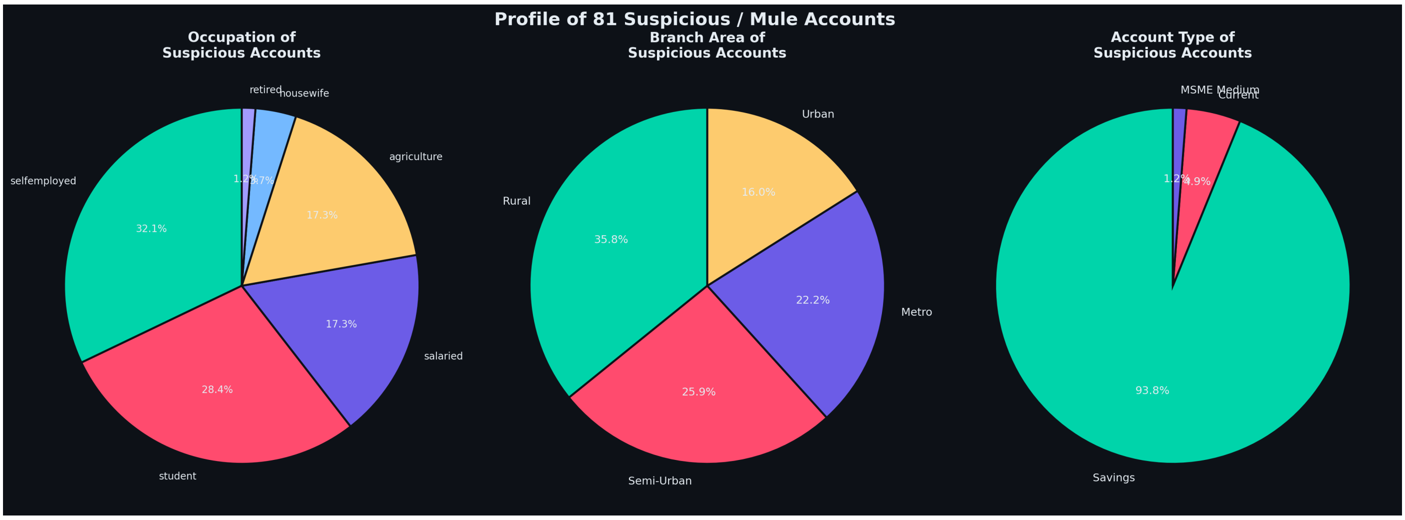
- **ALL top 20 have positive correlation -** suspicious accounts have higher values
- **F2506/F2507** tied at 0.1845 - likely same metric computed differently
- Max correlation is only **0.18 -** no single feature is a strong standalone predictor
- Confirms we need **non-linear models** (tree-based) to capture interactions
- The 18 features from problem statement are NOT in top 20 - data-driven features differ

7. Feature Block Composition



- **F0-499:** Proportions (yellow) and small floats (blue) - behavioral ratios
- **F500-999:** Heavy in constants (gray) - many inactive product flags
- **F1000-1499:** Binary flags + large floats - product holdings and amounts
- **F2500-2999:** Almost entirely **negative-capable floats** - the change/delta block, most predictive
- **F3500-3999:** Mixed - metadata, categoricals, and alert flags

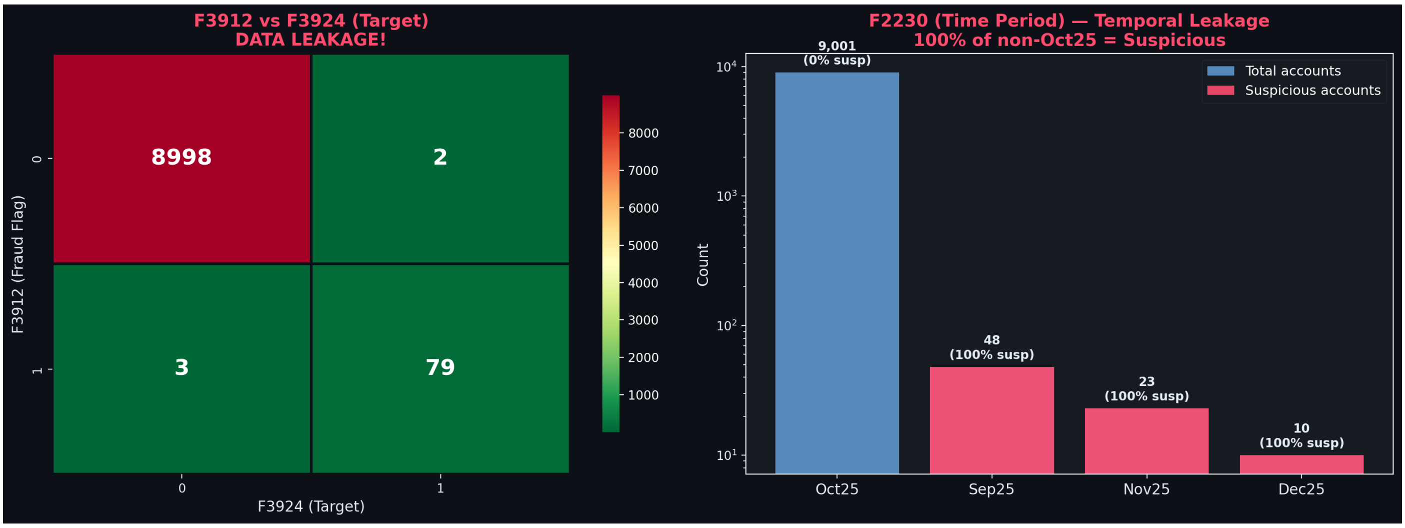
8. Suspicious Account Profile



The typical mule account is:

- A **self-employed (32%)** or **student (28%)** individual
- From a **Rural (36%)** or **Semi-Urban (26%)** branch
- With a **Savings account (94%)**
- Maps to real-world pattern: young individuals in smaller towns recruited as mule account holders

9. Data Leakage - Critical Finding

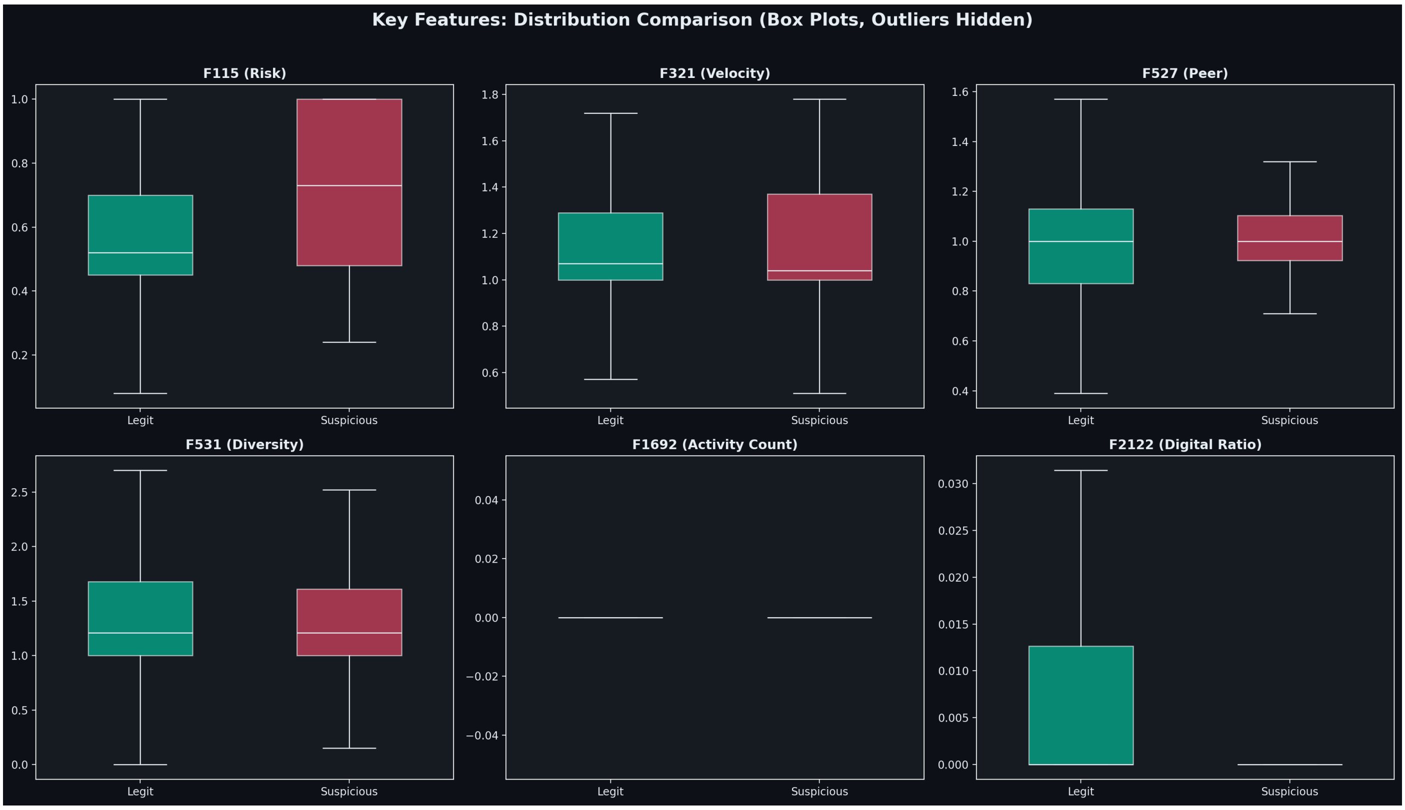


CAUTION: Two features MUST be excluded from all models: F3912 and F2230.

F3912 (Fraud Flag): The heatmap shows 79 of 82 F3912=1 accounts are also F3924=1. Only 5 mismatches across the entire dataset. Including this would create a model that simply copies this flag.

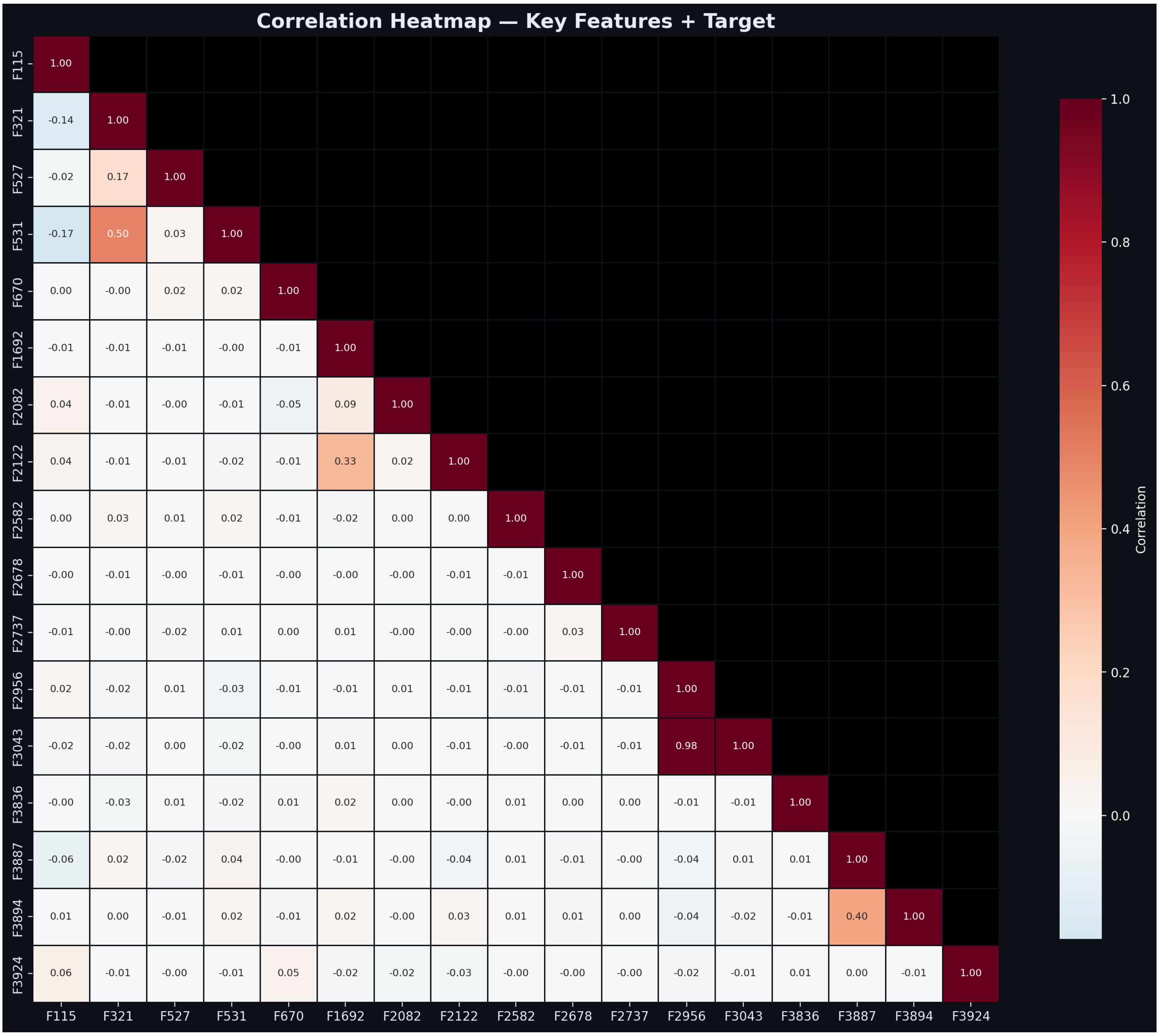
F2230 (Time Period): 100% of non-Oct25 accounts are suspicious (48+23+10 = 81). 0% of Oct25 accounts are suspicious. This is a data collection artifact, not a real pattern.

10. Key Feature Box Plots



- **F115 (Risk):** Median for suspicious is noticeably higher - clear upward shift
- **F321 (Velocity):** Similar medians but legit has wider spread
- **F531 (Diversity):** Legit accounts show wider whiskers - suspicious lack diversity
- **F2122 (Digital):** Strikingly different - suspicious median is at ~0

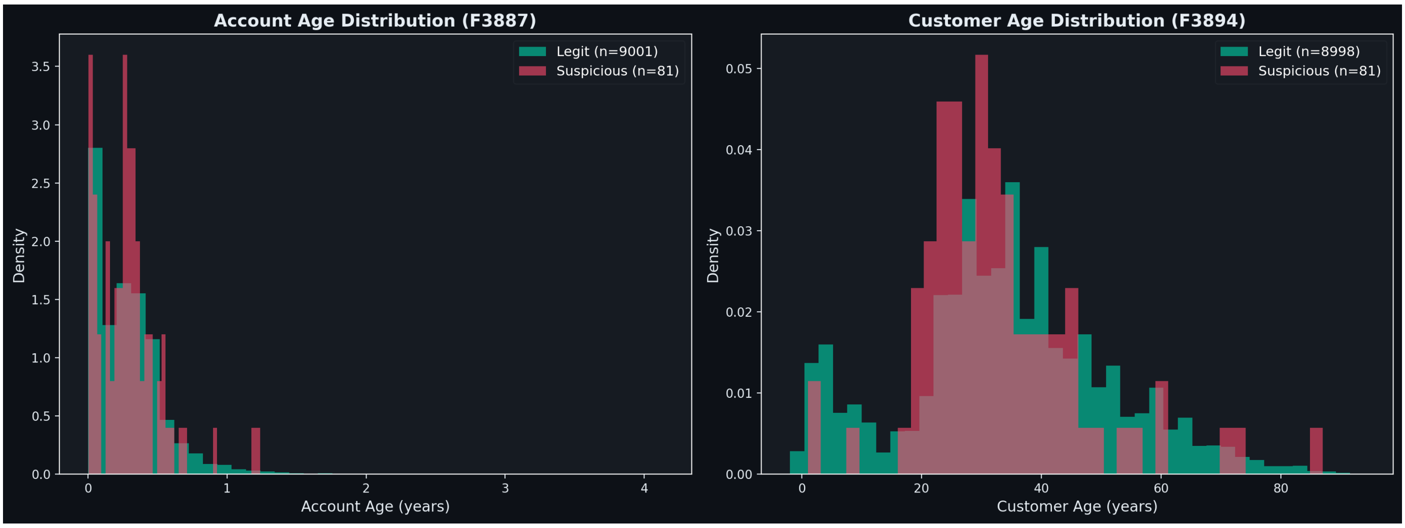
11. Key Feature Correlation Heatmap



- **F2956 / F3043 = 0.98** - near-duplicate features
- **F321 / F531 = 0.50** - velocity and diversity are moderately linked
- **F3887 / F3894 = 0.40** - account age and customer age correlated

TIP: Most features are near-zero correlated with each other - they capture independent signals. Excellent for model performance (low multicollinearity).

12. Age Distributions



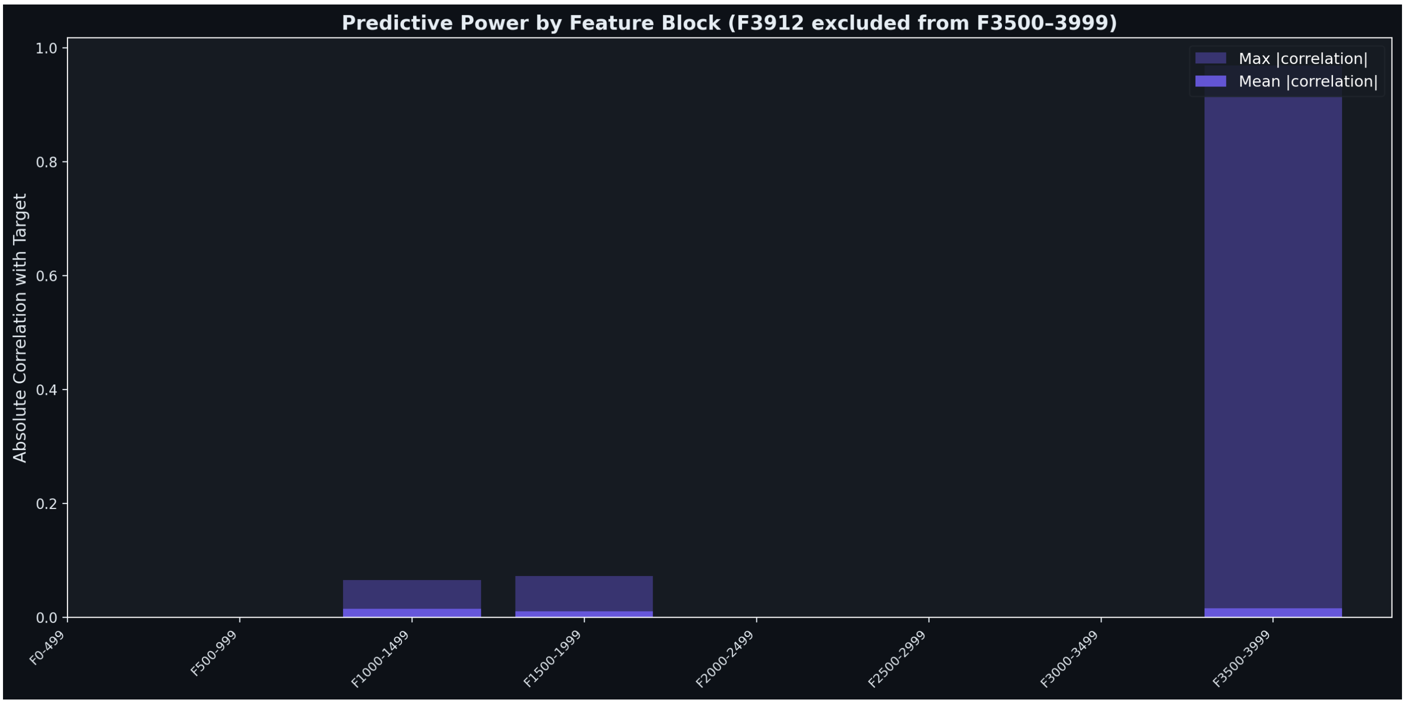
Account Age (F3887):

- Both classes heavily right-skewed - most accounts are **< 6 months old**
- Suspicious accounts concentrated in the **0-3 month range** - newly opened

Customer Age (F3894):

- Suspicious accounts peak sharply at **25-35 years**
- Legitimate accounts have a broader distribution extending to 70+

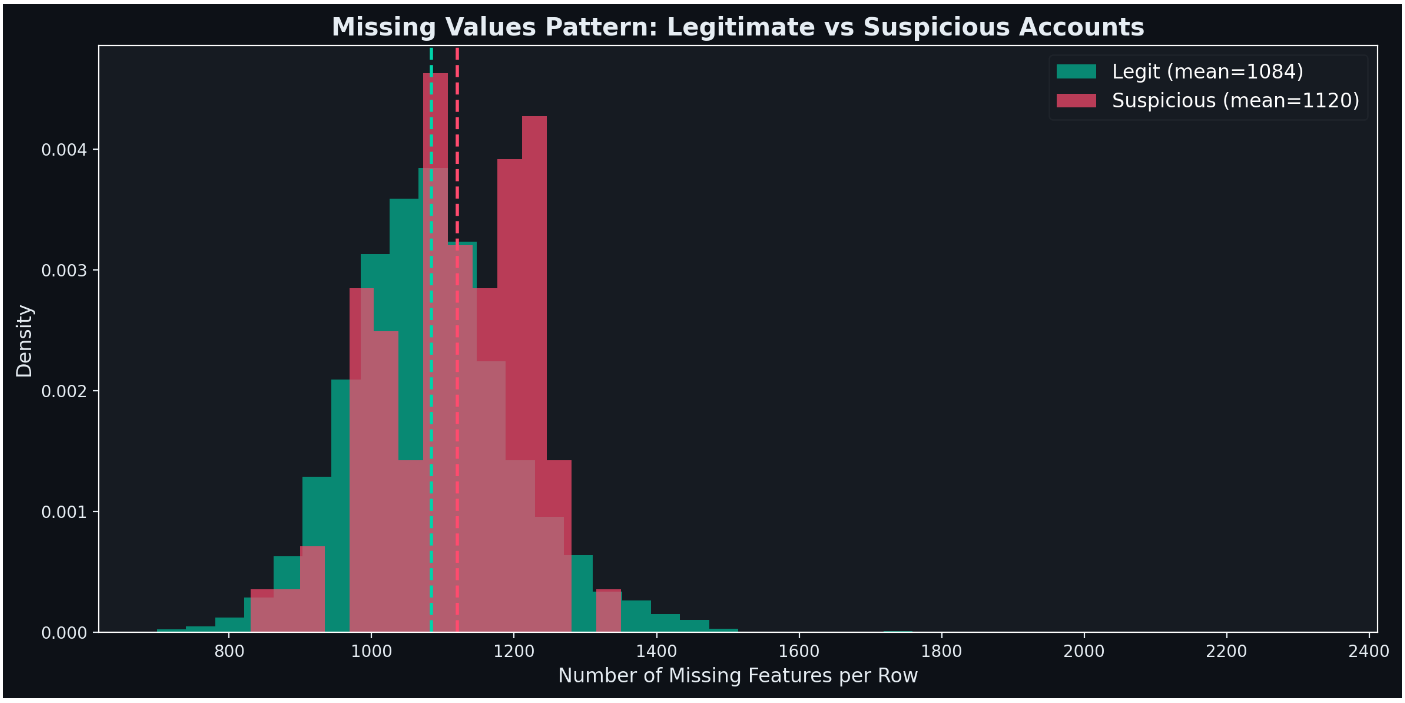
13. Block-Level Importance



Shows mean and max absolute correlation with target by 500-feature block.

- Delta/change blocks (F2500-F2999) show the **highest predictive signal**

14. Missing Values by Class



- Legit accounts: mean **1,084** missing features
- Suspicious accounts: mean **1,120** missing features - slightly higher

- Missingness is **not a strong class signal** but could provide marginal value

Summary of Key Visual Insights

Finding	Visual Evidence	Impact on Model
111:1 class imbalance	Graph 1	Must use SMOTE/class weights
F3912 data leakage	Graph 9 (left)	DROP - 0.97 corr with target
F2230 temporal leakage	Graph 9 (right)	DROP - 100%/0% temporal split
Students = highest risk	Graph 4	Occupation is a strong feature
Rural = highest risk area	Graph 4	Branch area adds predictive value
Suspicious = low digital	Graph 5, 10	F2122 is one of the best features
Suspicious = newer accts	Graph 12	Account age (F3887) is predictive
Delta features = best	Graph 6, 7	F2500-F2999 block is the gold mine
Features mostly independent	Graph 11	Low multicollinearity = good for trees
Missingness is structural	Graph 2, 14	Not class-dependent, use tree models

NOTE: All 14 graphs are saved permanently in files/graphs/ for presentation and documentation purposes.